

# An Analysis of Ridge regression

## Ridge regression

$$L(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|^2$$

$$\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$$

### Ridge regression -- Part 1

$$\hat{\beta} = (X^T X + \lambda I)^{-1} X^T Y \quad \{\displaystyle \hat{\beta} = \left( \mathbf{X}^T \mathbf{X} + \lambda \mathbf{I} \right)^{-1} \mathbf{X}^T \mathbf{Y} \}$$

### Ridge regression -- Part 2

with the "regularisation matrix"  $B = (A^T A + \Gamma^T \Gamma)^{-1} A^T A$   $\{\displaystyle B = \left( A^T A + \Gamma^T \Gamma \right)^{-1} A^T A \}$ . If the parameter fit comes with a covariance matrix of the estimated parameter uncertainties  $V_0$   $\{\displaystyle V_0\}$ , then the regularisation matrix will be

### Ridge regression -- Part 3

History Tikhonov regularization was invented independently in many different contexts. It became widely known through its application to integral equations in the works of Andrey Tikhonov and David L. Phillips. Some authors use the term Tikhonov–Phillips regularization. The finite-dimensional case was expounded by Arthur E. Hoerl, who took a statistical approach, and by Manus Foster, who interpreted this method as a Wiener–Kolmogorov (Kriging) filter. Following Hoerl, it is known in the statistical literature as ridge regression, named after ridge analysis ("ridge" refers to the path from the constrained maximum).

### Ridge regression -- Part 4

and is zero elsewhere. This demonstrates the effect of the Tikhonov parameter on the condition number of the regularized problem. For the generalized case, a similar representation can be derived using a generalized singular-value decomposition. Finally, it is related to the Wiener filter:

### Ridge regression -- Part 5

Regularization in Hilbert space Typically discrete linear ill-conditioned problems result from discretization of integral equations, and one can formulate a Tikhonov regularization in

the original infinite-dimensional context. In the above we can interpret  $A$   $\{\displaystyle A\}$  as a compact operator on Hilbert spaces, and  $x$   $\{\displaystyle x\}$  and  $b$   $\{\displaystyle b\}$  as elements in the domain and range of  $A$   $\{\displaystyle A\}$ . The operator  $A^* A + \Gamma^T \Gamma$   $\{\displaystyle A^* A + \Gamma^T \Gamma\}$  is then a self-adjoint bounded invertible operator.

### Ridge regression -- Part 6

$$E \|\hat{\beta} - \beta\|^2 \geq E \|\hat{\beta}_0\|^2 \geq \frac{\text{tr}(\Omega)}{n} \quad \{\displaystyle E \|\hat{\beta} - \beta\|^2 \geq E \|\hat{\beta}_0\|^2 \geq \frac{\text{tr}(\Omega)}{n} \}$$

### Ridge regression -- Part 7

$$\lambda^* := \frac{\text{tr}(\Omega)}{n} \quad \{\displaystyle \lambda^* := \frac{\text{tr}(\Omega)}{n} \}$$

### Ridge regression -- Part 8

Generalized Tikhonov regularization For general multivariate normal distributions for  $x$   $\{\displaystyle \mathbf{x}\}$  and the data error, one can apply a transformation of the variables to reduce to the case above. Equivalently, one can seek an  $x$   $\{\displaystyle \mathbf{x}\}$  to minimize

### Ridge regression -- Part 9

Lavrentyev regularization In some situations, one can avoid using the transpose  $A^T$   $\{\displaystyle A^T\}$ , as proposed by Mikhail Lavrentyev. For example, if  $A$   $\{\displaystyle A\}$  is symmetric positive definite, i.e.  $A = A^T > 0$   $\{\displaystyle A = A^T > 0\}$ , so is its inverse  $A^{-1}$   $\{\displaystyle A^{-1}\}$ , which can thus be used to set up the weighted norm squared  $\|\mathbf{x}\|_P^2 = \mathbf{x}^T A^{-1} \mathbf{x}$   $\{\displaystyle \|\mathbf{x}\|_P^2 = \mathbf{x}^T A^{-1} \mathbf{x}\}$  in the generalized Tikhonov regularization, leading to minimizing

### Ridge regression -- Part 10

where  $A = \begin{pmatrix} A & \Gamma \end{pmatrix}$   $\{\displaystyle A = \begin{pmatrix} A & \Gamma \end{pmatrix}\}$  and  $b = \begin{pmatrix} b \\ 0 \end{pmatrix}$   $\{\displaystyle b = \begin{pmatrix} b \\ 0 \end{pmatrix}\}$ , for some suitably chosen Tikhonov matrix  $\Gamma$   $\{\displaystyle \Gamma\}$ . In many cases, this

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matrix is chosen as a scalar multiple of the identity matrix ( $\Gamma = \alpha I$ ), giving preference to solutions with smaller norms; this is known as L2 regularization. In other cases, high-pass operators (e.g., a difference operator or a weighted Fourier operator) may be used to enforce smoothness if the underlying vector is believed to be mostly continuous. This regularization improves the conditioning of the problem, thus enabling a direct numerical solution. Treating it as an ordinary least squares problem with augmented matrices  $\mathcal{A}$  and  $\mathcal{b}$ , the solution is

### Ridge regression -- Part 11

Relation to probabilistic formulation The probabilistic formulation of an inverse problem introduces (when all uncertainties are Gaussian) a covariance matrix  $C_M$  representing the a priori uncertainties on the model parameters, and a covariance matrix  $C_D$  representing the uncertainties on the observed parameters. In the special case when these two matrices are diagonal and isotropic,  $C_M = \sigma_M^2 I$  and  $C_D = \sigma_D^2 I$ , and, in this case, the equations of inverse theory reduce to the equations above, with  $\alpha = \sigma_D^2 / \sigma_M^2$ .

### Ridge regression -- Part 12

where the expectations treat  $X$  as fixed and  $Y'$  is test response data, independent from  $Y$  (and hence independent from the estimators  $\hat{\beta}_0$  and  $\hat{\beta}_{\lambda^*}$ ). Of course, in practice the formula for  $\lambda^*$  is used by plugging in statistical estimators for the unknown parameters  $\beta$  and  $\varsigma^2$ . When  $n > p$ , the most natural estimators for these parameters are the usual least-squares ones:

### Ridge regression -- Part 13

$$\|Ax - b\|^2 + \|\Gamma x\|^2 = \|Ax - b\|^2, \quad \left\| \begin{bmatrix} A \\ \Gamma \end{bmatrix} x - \begin{bmatrix} b \\ 0 \end{bmatrix} \right\|^2 = \left\| \begin{bmatrix} A \\ 0 \end{bmatrix} x - \begin{bmatrix} b \\ 0 \end{bmatrix} \right\|^2,$$

### Ridge regression -- Part 14

where  $x$  and  $b$  may be of different sizes and  $A$  may even be non-square. The standard approach is ordinary least squares linear regression. However, if no  $x$  satisfies the equation or more than one  $x$  does—that is, the solution is not unique—the problem is said to be ill posed. In such cases, ordinary least squares estimation leads to an overdetermined, or more often an underdetermined system of equations. Most real-world phenomena have the effect of low-pass filters in the forward direction where  $A$  maps  $x$  to  $b$ . Therefore, in solving the inverse-problem, the inverse mapping operates as a high-pass filter that has the undesirable tendency of amplifying noise (eigenvalues / singular values are largest in the reverse mapping where they were smallest in the forward mapping). In addition, ordinary least squares implicitly nullifies every element of the reconstructed version of  $x$  that is in the null-space of  $A$ , rather than allowing for a model to be used as a prior for  $x$ . Ordinary least squares seeks to minimize the sum of squared residuals, which can be compactly written as